

Infrastructure Sizing & Deployment Strategy

Date: June 2026 | Project: Quantum Terminal Pro

1. Optimal Server Location

Based on comprehensive empirical testing and latency benchmarks, it has been conclusively established that the optimal hosting infrastructure locations for the platform are either the **Netherlands** or **Germany**. Deploying in these regions guarantees minimal round-trip time (RTT) for websocket data feeds and ensures consistent uptime across major digital asset exchanges.

2. Monitoring Optimization for Evaluation

All diagnostic utilities, platform logging depth, and telemetry monitoring configurations have been dialed down to their absolute minimum thresholds. This explicit tuning has been performed to facilitate testing on low-power, entry-level servers, removing unnecessary resource allocation and performance bottlenecks during the trial or baseline assessment phases.

3. Capacity & Baseline Resource Configurations

Under these optimized low-overhead monitoring environments, rigorous performance testing confirms that the architecture can comfortably sustain concurrent service for ****100 to 150 active users****. The validated system profile requires the following baseline hardware specifications:

Hardware Resource	Minimum Recommended Specification
CPU Capacity	2-Core Processor (Standard/Medium performance profile tier)
Physical Memory (RAM)	2 GB of Physical RAM
Virtual Memory (Swap)	15 to 20 GB of configured Swap Space (paging file utilized to partition active memory and disk storage safely under peak load conditions)

Summary Assessment: The combination of 2 GB physical RAM paired with 15–20 GB of high-speed swap space ensures adequate memory allocation overhead, preventing unexpected out-of-memory errors while serving up to 150 concurrent platform clients.